

Transmission Control Protocol

Module 06

Duration: 27:00

Video: <https://www.youtube.com/watch?v=9bKXrX4lz6g>

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0:00 [Music]

0:15 Hello friends I am Professor yogesh Chaba from Department of computer science and engineering Guru jambheshwar

0:22 University of Science and Technology Hisar harana I welcome you all to the lecture series of of cyber and

0:31 information security in this lecture we will study about transmission control

0:37 protocol which is used for connection oriented transmission at transport layer

0:44 level for end to end connectivity of computers first of all please see the contents which I shall

0:52 cover after covering brief introduction about TCP I shall take up header of TCP

1:00 which will be followed by connection management and flow control process in

1:07 TCP so now let us start the foundation of TCP was laid in May

1:15 1974 when Institute of electrical and electronic Engineers itle published a

1:23 paper entitled a protocol for packet Network intercommunication the papers author

1:31 Surf and KH described an internetworking protocol for sharing resources using

1:40 packet switching among the nodes a central control component of this model

1:46 was the transmission control program TCP that incorporated both connection oriented

1:55 links and datagram services between host this transmission control program was

2:04 later divided into a modular architecture consisting of the transmission control protocol at the

2:12 connection oriented layer and the Internet Protocol at the internet

2:19 layer the model became known informally as PCP IP although formerly it was

2:27 henceforth called the Internet Protocol suit PCP is a reliable stream delivery

2:37 service that guarantees that all bytes received will be identical with bytes

2:44 sent and in the correct order this fundamental technique requires the receiver to respond with an

2:52 acknowledgement message as it receives the data the sender keeps a record of
3:00 each packet it sends the sender also maintains a timer from when the packet
3:07 was sent and retransmits a packet if the timer expires before the message has
3:15 been acknowledged the timer is needed in case a packet gets lost or corrupted TCP
was
3:25 formally defined in RFC 793 as time went on various errors and
3:33 inconsistencies were detected and the requirements were changed in some
3:39 areas these clarifications and some bugs fixes are detailed in RFC
3:48 112 extensions are given in RFC 1323 while IP handles actual delivery of
3:58 the data TCP keeps track of the individual units of data transmission
4:04 called segments the internet layer encapsulates each TCP segment into an IP
4:13 packet by adding an IP header that includes the destination IP address when
4:21 the client program on the destination computer receives them the TCP layer
reassembles the individual visual
4:30 segments and ensures they are correctly ordered and error
4:36 free for lack of a better term we will use acronym tpdu that is transport protocol
data
4:46 unit for messages sent from transport layer thus tpd exchanged by the
4:55 transport layer are contained in packets at the Network layer in turn packets are
5:04 contained in frames at the data link layer this nesting is Illustrated in
5:13 figure TCP service is obtained by the sender and receiver creating end points
called
5:21 sockets each socket has a socket number consisting of the IP address of the host
and a 16bit number
5:30 local to that host called a port a port is the TCP name for a transport service
5:37 access point tsap a socket may be used for multiple connections at the same time in
other
5:47 words two or more connections May terminate at the same socket connections are
identified by the
5:56 socket identifiers at both ends that is Socket 1 socket
6:04 2 no virtual circuit numbers or other identifiers are used port numbers below
6:13 1024 are called well-known ports and are reserved for standard
6:20 Services a few of the better known ones are listed in table the internet
6:27 assigned numbers Authority are AA is responsible for maintaining the official
6:34 assignments of port numbers for specific uses as we can see in table Port 20
6:42 supports TCP and UDP communication for FTP data
6:50 transfer Port 23 supports TCP and UDP communication for tnet protocol
7:00 Port 25 supports TCP and UDP communication for simple mail transfer

7:08 protocol SMTP in email system Port 53 supports TCP and UDP
7:18 communication for domain name system DNS Port 69 is for trial file transfer
7:28 protocol tftp Port 0 supports TCP sctp and UDP
7:38 communication for hyper text transfer protocol
7:44 HTTP Port 107 supports TCP and UDP for remote tet service
7:54 protocol Port 109 supports TCP and UDP P for post office protocol
8:03 V2 Port 110 supports TCP and UDP for post office protocol
8:11 V3 Port 115 supports TCP for simple file transfer
8:20 protocol Port 119 supports TCP for Network news transfer protocol nnt
8:30 P Port 143 supports TCP for internet message access protocol
8:39 IMAP Port 220 supports TCP and UDP for
8:44 internet message access protocol IMAP Port 500 supports TCP and UDP for
8:54 Internet Security Association and Key Management prot
9:00 protocol Port 530 supports TCP and UDP for remote procedure call
9:10 RPC Port 554 supports TCP and UDP for realtime streaming protocol
9:20 rtsp protocol 587 supports TCP for email message submission SMTP
9:29 Port 698 supports UDP for optimized link state routing
9:37 osr Port 953 supports TCP and UDP for
9:43 domain name system R and DC Service Port 995 supports TCP for post
9:53 office protocol three connections are full duplex and point to point
9:59 full duplex means that traffic can go in both directions at the same time point
10:06 to point means that in each connection has exactly two end points TCP does not
10:15 support multicasting or broadcasting a TCP connection is a bite stream not a
message
10:23 stream message boundaries are not preserved end to end for for example if
10:30 the sending process does four 512 byte right to a TCP stream these data may be
10:38 delivered to the receiving process as four 512 byte chunks two 1024 by chunks
10:47 or one 2048 byte chunk as shown in figure or some other way
10:55 also there is no way for the receiver to detect the the unit in which the data were
11:03 written figure a shows four 512 by segments sent as separate IP datagram
11:11 and figure B shows the 2048 byte of data delivered to the application in a single
11:19 read call after basic of TCP and Port let us
11:26 see the TCP segment header figure shows the layout of a TCP segment
11:35 every segment begins with a fixed format of 20 by header the fixed header may be
11:42 followed by header options after the options if any up to
11:49 65535 -20 -20 which is equal to
11:57 65495 databytes may follow where the first 20 refers to

12:04 the IP header and second 20 refers to the TCP header segments without any data are

12:13 legal and commonly used for acknowledgements and control messages now let us discuss the TCP

12:22 header Fields One By One The Source port and destination Port

12:28 fields identify the local end points of the connection a port plus its host IP

12:36 address forms a 48 bit unique end point the source and destination end

12:44 points together identify the connection the sequence number and acknowledgement number field performs

12:53 their usual functions for the purpose of delivery confirmation no note that the

13:00 acknowledgement number specifies The Next Bite expected not the last bite

13:06 correctly received both are 32 bits long because every bite of data is numbered

13:14 in a TCP stream the TCP header length tells how

13:20 many 32bit words are contained in the TCP header this information is needed

13:28 because the op options field is of variable length technically this field really indicates the start of the data

13:36 within the segment measured in 32bit words but that number is just the header

13:44 length in words so the effect is the same next comes a six-bit field that is

13:52 not used next are six 1 bit Flags urg is set

13:59 to one if the Urgent pointer is in use the Urgent pointer is used to indicate a

14:07 bite offset from the current sequence number at which urgent data are to be

14:15 found the AK bit is set to one to indicate that the acknowledgement number is

14:23 valid if AK is zero the segment does not contain an acknowledgement so the acknowledgement number field is

14:32 ignored the psh bit indicates pushed data the receiver is hereby kindly

14:40 requested to deliver the data to the application upon arrival and not buffer

14:46 it until a full buffer has been received the rst bit is used to reset a

14:53 connection that has become confused due to host crash or some mother

15:01 15 minutes, 1 second reason the sin bit is used to establish connections the connection request has

15:08 sin as one and akk as zero to indicate that the piggy back acknowledgement

15:16 field is not in use the connection reply does bear an

15:23 acknowledgement so it has sin as one and AK as one

15:29 in essence the sin bit is used to denote connection request and connection

15:37 accepted if AK bit is used to distinguish between those two

15:45 possibilities the fin bit is used to release a connection it specifies that

15:52 the sender has no more data to transmit flow control in TC p is handled

16:00 using a variable sized sliding window the window size field tells how many

16:07 bytes may be sent starting at the byte acknowledged a check sum is also
16:14 provided for extra reliability it check sums the header the data and the conceptual
pseudo
16:25 header after TCP header let us move over to to socket Primitives in
16:31 TCP socket Primitives are widely used for internet programming they are listed in
16:40 figure the socket primitive creates a new communication end point newly created
socket do not have
16:50 Network addresses these are assigned using the bind primitive bind primitive is
used to
16:58 attach a local address to a socket next comes the listen call which
17:06 announces willingness to accept connections and allocate space to queue incoming
calls for the case that several
17:15 clients try to connect at the same time to block waiting for an incoming
17:23 connection the server executes and accept primitive now let us look at the client
17:31 side here to a socket must first be created using the socket primitive but
17:37 bind is not required since the address used does not matter to the
17:45 server the connect primitive blocks the caller and actively starts the connection
17:52 process when the appropriate tpdu is received from the server the client process is
unblocked and the connection
18:01 18 minutes, 1 second is established both sides can now use send or receive to
transmit and receive data
18:11 over the full duplex connection connection release with
18:18 socket is symmetric when both sides have executed a close primitive the
connection is
18:27 released now we shall learn how TCP connection is established connections are
established
18:36 in TCP by means of the three way handshake process to establish a
18:42 connection one side say the server passively waits for an incoming Connection by
executing the listen and
18:51 accept Primitives the other side say the client executes a connect PR
18:59 primitive the connect primitive sets a TCP segment with a sin bit on and AK bit
19:07 off and waits for a response when this segment arrives at
19:15 the destination the TCP entity checks to see if there is a process that has done
19:22 a listen on the port given in the destination Port field if not it sends a
19:29 reply with the RSD bit on to reject the connection if some process is listening
19:37 to the port that process is given the incoming TCP segment it can then either
accept or
19:46 reject the connection if it accepts an acknowledgement segment is sent

19:53 back the sequence of TCP segments sent in the normal case is shown in figure a
20:01 20 minutes, 1 second on the left side on receipt of sin with SQ as X host
20:09 to return sin with seq as Y and AK as $x +$
20:17 1 further host one sends data with seq
20:22 as $x + 1$ and AK as $y + 1$
20:30 in the event that Two Hosts simultaneously attempt to establish a connection
between the same two sockets
20:39 the sequence of events is as Illustrated in Part B on right side of
20:48 figure which clearly shows that Collision will occur after connection setup let us
see
20:57 how TCP connection is released to release a connection either
21:04 party can send a TCP segment with a fin bit set which means that it has no more
21:12 data to transmit when the fin is acknowledged that direction is shut down
21:18 for new data data may continue to flow indefinitely in the other
21:26 direction when both directions have been shut down the connection is released
21:32 normally four TCP segments are needed to release a
21:39 connection one Fin and one AK for each Direction however it is possible for the
21:49 first AK and the second fin to be contained in the same segment reducing
21:56 the total segment count to three to avoid the two Army problem
22:04 timers are used if a response to a fin is not forthcoming within two maximum
22:12 packet lifetime the sender of the fin releases the connection the other side will
22:21 eventually notice that nobody seems to be listening to it anymore and will time
22:27 out as as well now let us see TCP connection management and flow
22:34 control as mentioned earlier window Management in TCP is not directly tied to
22:42 acknowledgements as it is in most data link protocols now let us see TCP
22:49 connection management and flow control process step by
22:55 step in Step One the receiver has a 4,096 byte buffer as shown in right side
23:04 of figure if the sender transmits a 2048 byte segment in step one with sequence
23:14 as zero and it is correctly received the receiver will acknowledge the segment in
23:21 step two however since receiver has now only 2048 bytes of buffer space it will
23:30 advertise a window of 2048 starting at the next byte expected and AK as
23:40 2048 at this point 2K space is available in receiver buffer as shown on right
23:48 side in step two in step three the sender transmits
23:55 another 2048 bytes with sequence as 2048 receiver receives this 2K data and
24:04 it receiver buffer is full so in step four it is acknowledged
24:11 with AK as 4096 and win as zero the sender must stop until the
24:21 application process on the receiving host has removed some data from the

24:27 buffer at which TCP can advertise a larger window when the window is zero the
24:35 cender may not normally send segments with two exceptions first Urgent data may
be sent
24:44 for example to allow the user to kill the process running on the remote
24:51 machine second the sender may send a one byte segment to make the receiver will
24:58 re announce the next bite expected and window size the TCP standard explicitly
25:08 provides this option to prevent deadlock if a window announcement ever gets
25:15 lost as soon as receiver application reads 2K data it has 2K window size so
25:24 receiver replies with AK as 409 96 and win as
25:32 2048 in Next Step transmitter will send 1K data with sequence as
25:41 4096 so in this way communication takes place and endtoend flow control through
25:50 TCP so friends this was transmission control protocol finally summarizing
25:59 we learned about introduction to TCP protocol which was followed by TCP
26:06 header in the end we discussed connection management and flow control in
26:14 TCP hope the concepts explained in this lecture were understand and helpful hope
26:22 to see you in the next lecture till then goodbye enjoy the day thank you
26:32 [Music]